

VersaBank

Dynamic Holdback Model

Receivable Purchase Program — Credit Risk Framework

Work in Progress | May 22 – May 28, 2026

Prepared for: Tim

Project Background & Timeline

May 22	May 22–25	May 26	May 26–28
Meeting with Lauren Began organizing loan data and planning the holdback framework structure.	Scope Refinement Identified need to align model with VersaBank’s Receivable Purchase Program (RPP). General default model adjusted to reflect RPP-specific risk.	Meeting with Tim Reviewed early results. Identified key gaps: partner data, LGD assumption, and calibration of PD scores. Adjustments made based on feedback.	Ongoing Development Model training, validation, holdback calculation, and vintage analysis completed. Documentation and presentation prepared.

Approximately one week of active development. This is an early-stage framework — directionally sound but not yet production-ready.

The Problem We Are Solving

What is a holdback — and why does it need to be dynamic?

When VersaBank purchases loan receivables through its Receivable Purchase Program (RPP), it takes on the risk that some borrowers will stop paying. A holdback is the financial cushion held back from each lending partner to absorb those losses if they occur.

Current State	The Problem	The Solution
A flat holdback % is applied to all lending partners — regardless of the actual risk profile of the loans they originate.	Partners who originate riskier receivables are subsidized by those who originate safer ones. The bank may be under-reserved on high-risk portfolios.	Build a model that assigns a Probability of Default to every loan, then calculate a partner-specific holdback using the Expected Loss formula.

$$EL = PD \times EAD \times LGD$$

$$\text{Holdback \%} = (\text{Expected Loss} + \text{Variance Buffer}) / \text{Total Portfolio Outstanding}$$

Dataset Overview

Loan_default.csv — 255,347 loan records

255,347

Total Loan Records

18

Raw Features

11.6%

Portfolio Default Rate

0

Missing Values

Key Features Used in the Model

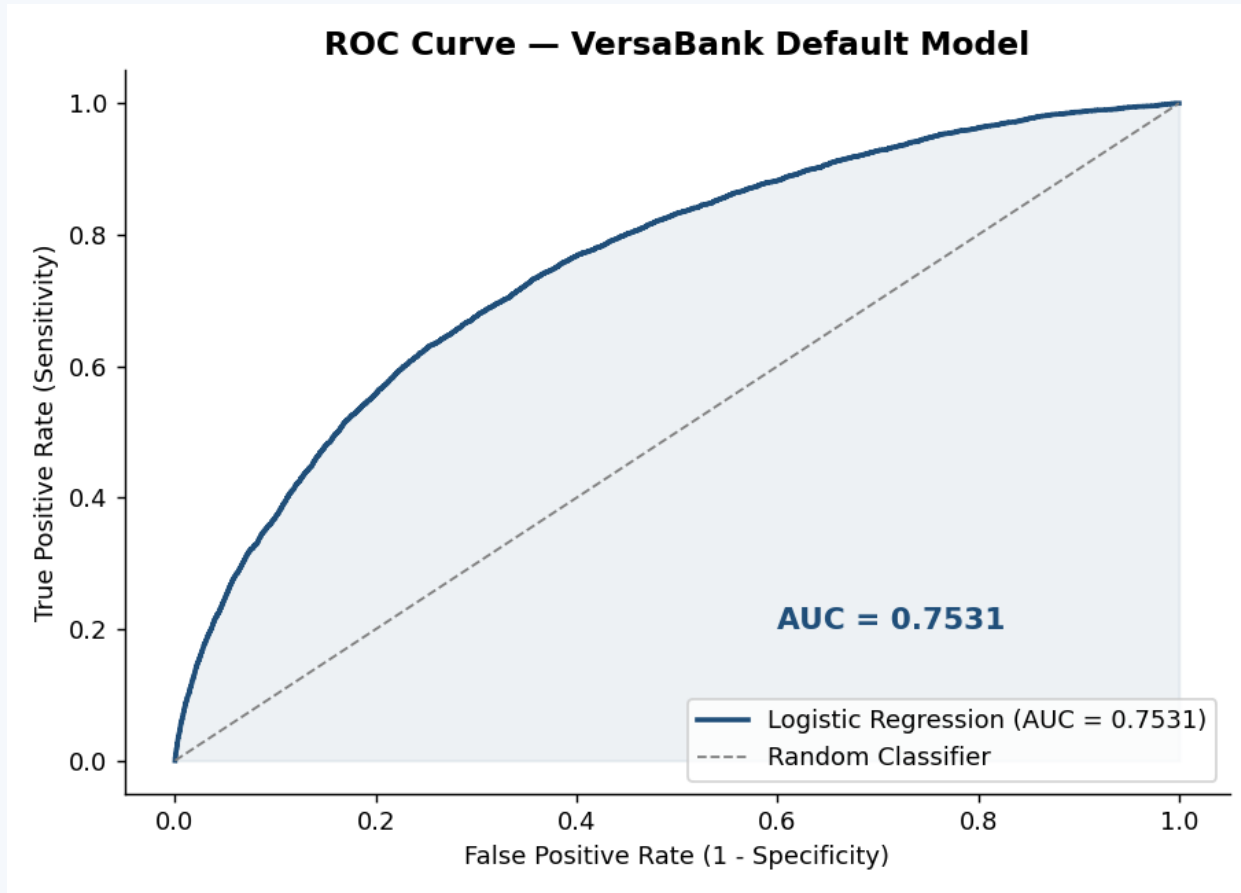
- Credit Score, Income, Loan Amount
- Interest Rate, DTI Ratio, Months Employed
- Employment Type, Loan Purpose, Education
- Has Mortgage, Has Co-Signer, Has Dependents
- Engineered: Lender Partner, Origination Date

Preprocessing Steps Applied

- Winsorization — outliers capped at 1st / 99th percentile
- One-hot encoding — text categories converted to 0 / 1
- Standardization — all continuous features scaled to mean = 0
- Binary encoding — Yes/No columns converted to 1 / 0
- Target variable: Default (0 = performing, 1 = defaulted)

Model Performance — ROC-AUC: 0.7531

Logistic Regression | 80/20 stratified train-test split | class_weight=balanced



What This Means

- ROC-AUC measures how well the model separates defaulting loans from performing ones
- 0.50 = coin flip | 1.00 = perfect
- Industry minimum for credit risk models: 0.70
- Our model: 0.7531 — clears the threshold
- Tested on 51,070 loans the model had never seen

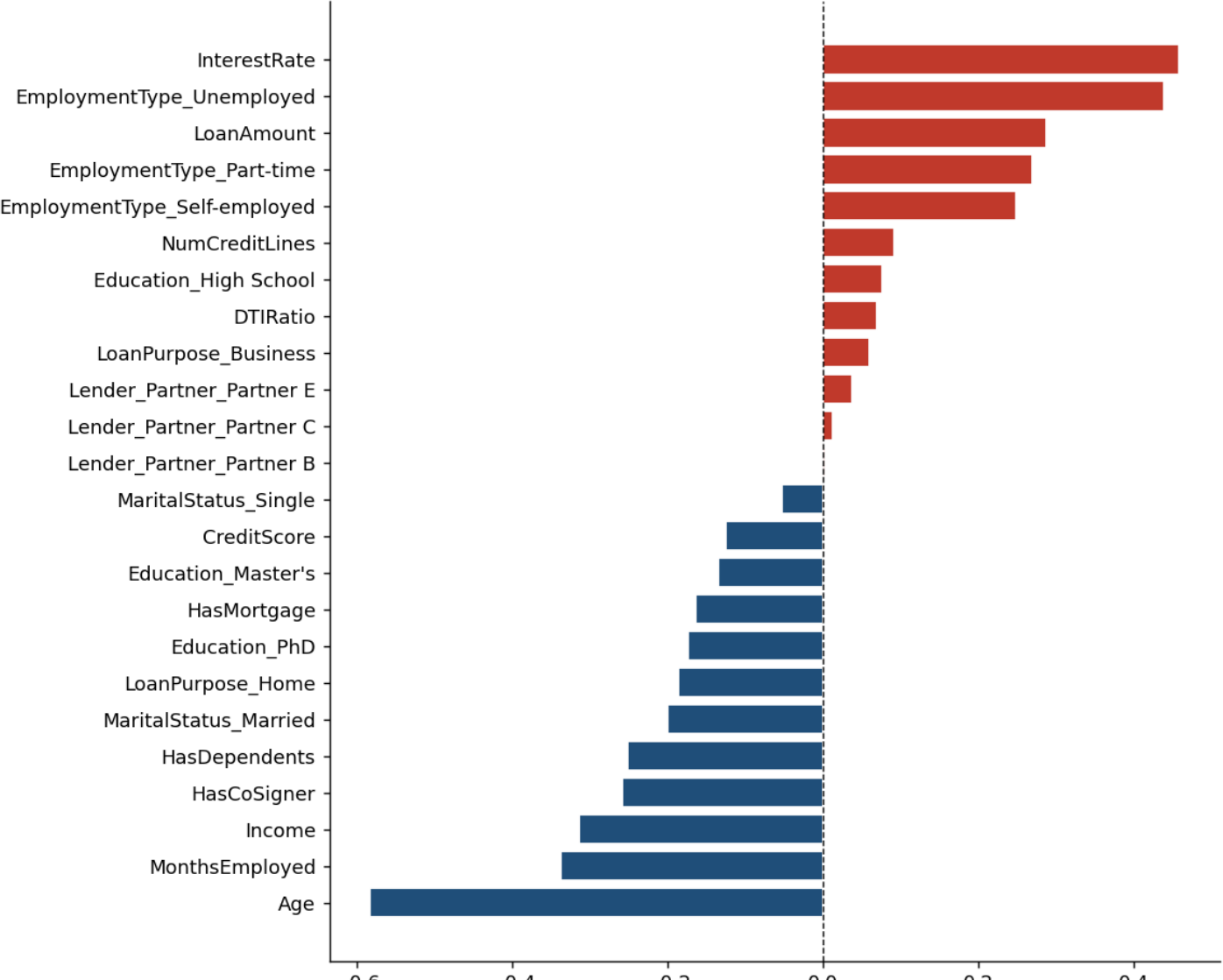
Pseudo $R^2 = 0.119$

The model explains ~12% of variation in defaults.
88% of what drives default is not yet captured
by this dataset — a known limitation at week 1.

Key Drivers of Default Risk

Logistic regression coefficients — all findings statistically significant ($p < 0.05$) unless noted

Top Predictors of Default Risk
(Red = increases risk | Blue = reduces risk)



Strongest Risk Amplifiers

- High interest rate (+0.46)
- Unemployed status (+0.44)
- Large loan amount (+0.30)
- Part-time employment (+0.27)

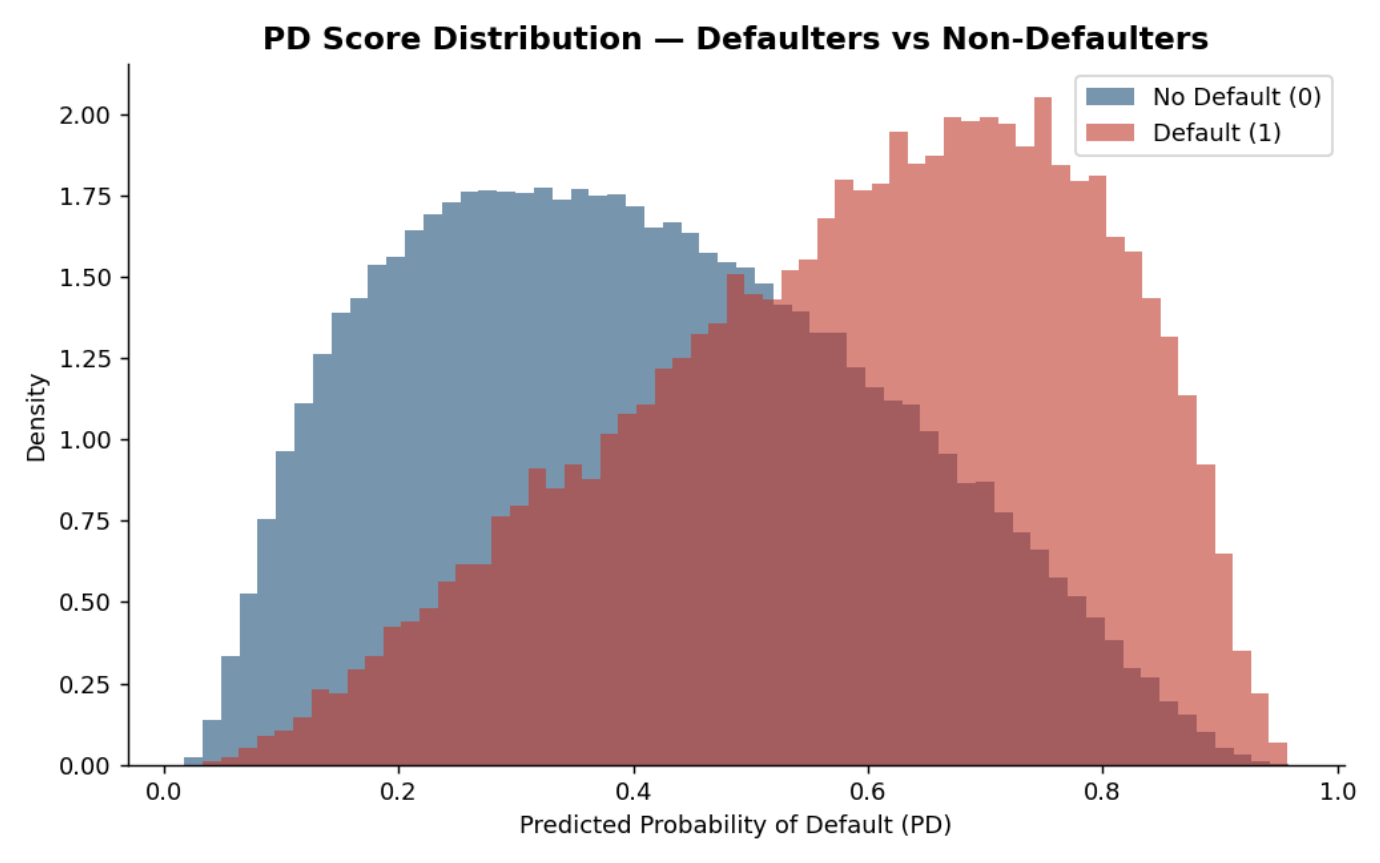
Strongest Protective Factors

- Older borrower age (−0.59)
- Longer employment history (−0.34)
- Higher income (−0.34)
- Co-signer present (−0.26)

Notable: Loan Term not significant ($p = 0.63$).
Loan Purpose signals are mixed and weak.

Probability of Default — Score Distribution

The model separates defaulters from non-defaulters, but scores require calibration



Calibration Issue

The model's average PD score is 43%.
The actual portfolio default rate is 11.6%.
The scores rank loans correctly but
overstate raw probabilities.

Score vs. Reality

Model PD Range	Actual Default Rate
0% – 11%	1.4%
11% – 21%	2.6%
39% – 49%	9.2%
77% – 86%	37.4%
86% – 96%	55.4%

Fix: Platt Scaling calibration — planned next step.

Holdback Calculation — $EL = PD \times EAD \times LGD$

Expected Loss per loan aggregated to partner level, plus a variance buffer for unexpected losses

$$EL = PD \times EAD \times LGD$$

$$\text{Holdback \%} = (\sum EL + \text{Variance Buffer}) / \text{Total EAD}$$

PD
Probability of Default

From logistic regression.
Estimated likelihood a
borrower stops paying.

EAD
Exposure at Default

Outstanding loan balance
at time of default.
We use original loan amount.

LGD
Loss Given Default

% of loan bank cannot
recover. Assumed 50%
(Basel II conservative floor).

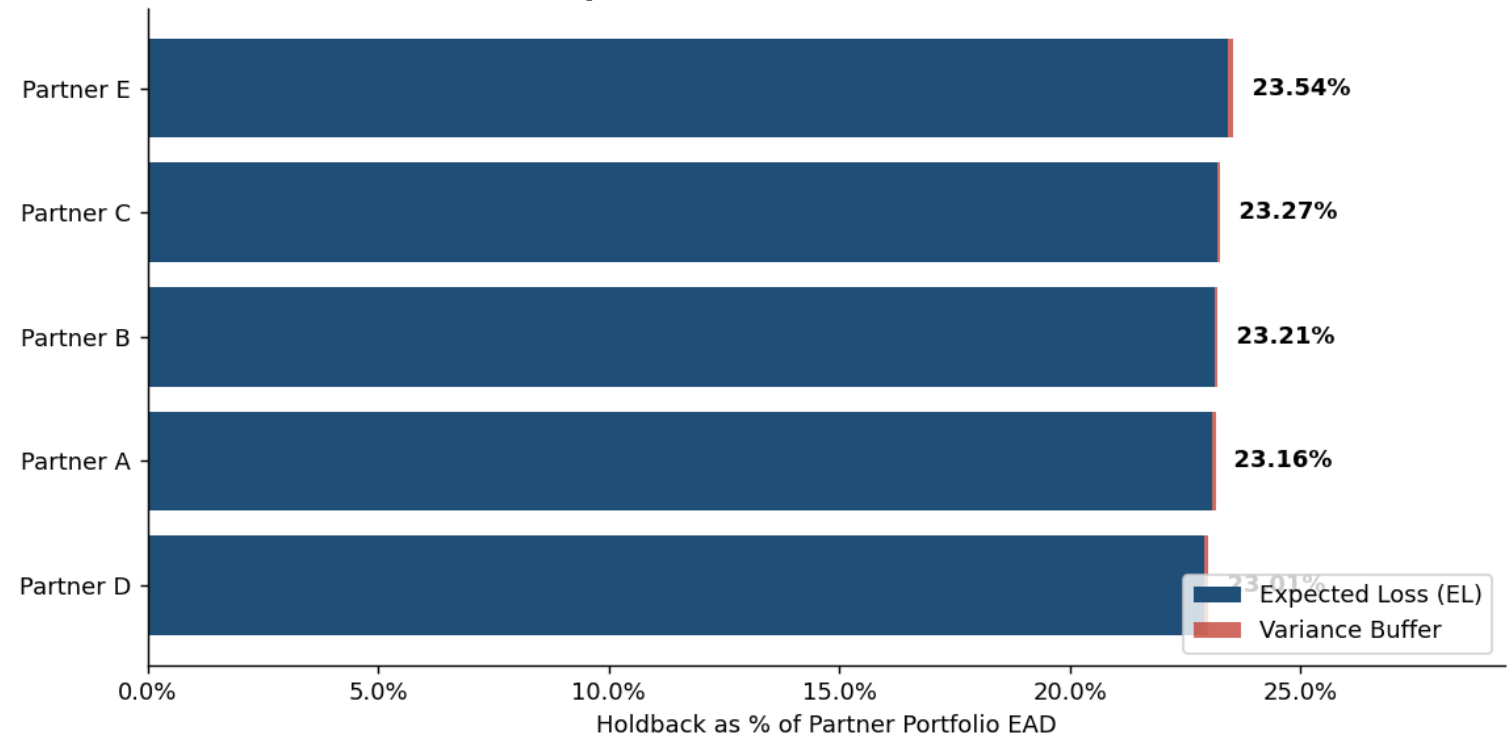
Variance Buffer

Added on top of Expected Loss to absorb unexpected macroeconomic shocks (e.g. a sudden recession or sector-wide job losses). Calculated as 1 portfolio-level standard deviation of loan-level EL per partner. This is consistent with Basel II’s distinction between Expected Loss (covered by pricing) and Unexpected Loss (covered by capital / holdback reserves).

Recommended Holdback by Lending Partner

EL + Variance Buffer as % of total partner portfolio outstanding

Recommended Holdback % by Lender Partner
(Expected Loss + Variance Buffer)



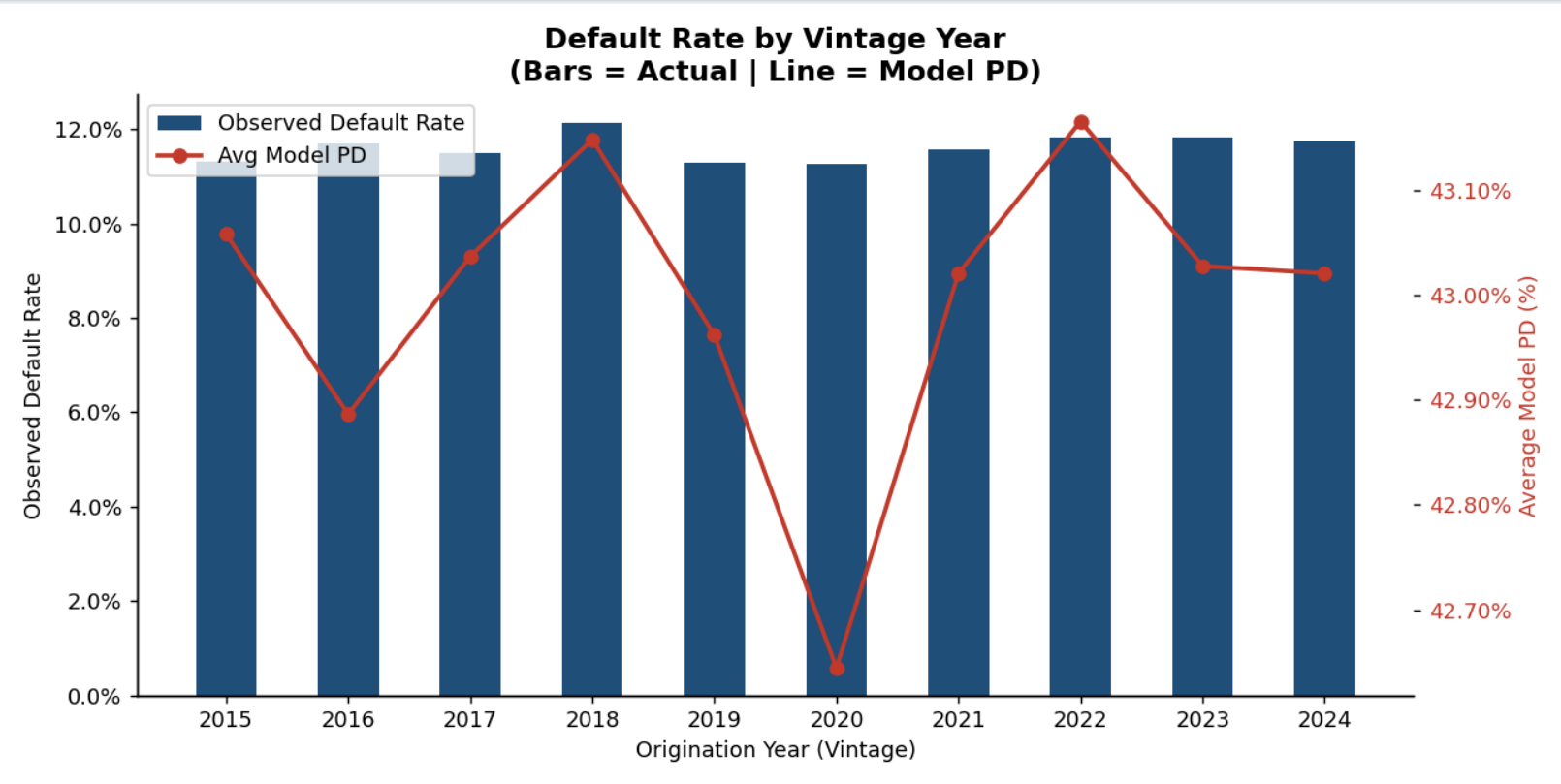
Partner	Total EAD	EL %	Buffer %	Holdback %
Partner E	\$3.25B	23.42%	0.12%	23.54%
Partner C	\$6.53B	23.19%	0.08%	23.28%
Partner B	\$8.17B	23.14%	0.07%	23.21%
Partner A	\$9.77B	23.10%	0.07%	23.17%
Partner D	\$4.86B	22.91%	0.10%	23.01%

Important caveat

The 0.53 pp spread between partners is narrow because partner labels are currently simulated. None of the partner variables are statistically significant ($p > 0.17$). Real RPP originator data will produce meaningful differentiation.

Vintage & Cohort Analysis — 10 Years of Originations

Grouping loans by origination year to identify which periods produced the highest default rates



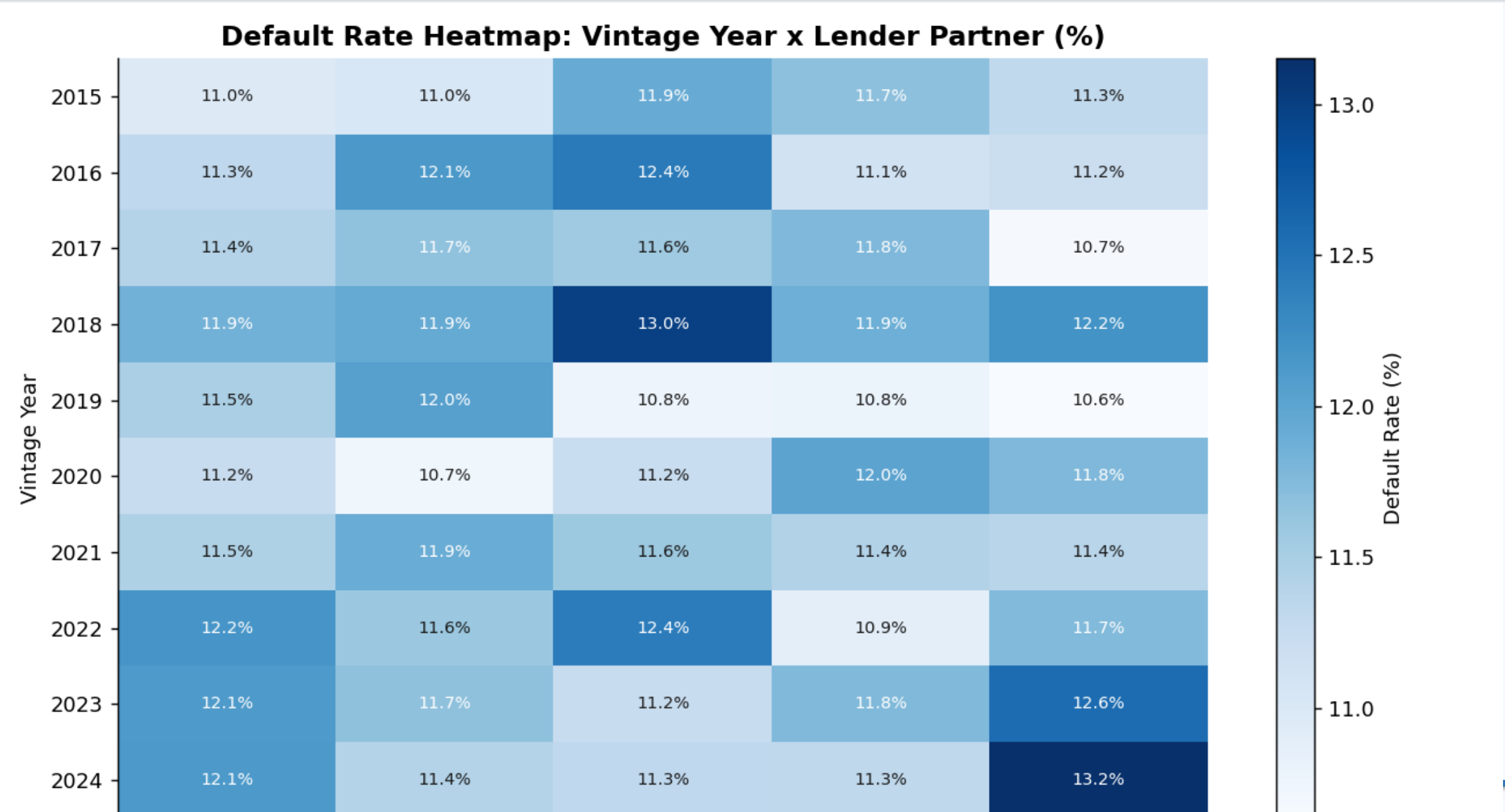
Key Findings

- 2018 vintage is the worst performing (12.1% default rate)
- 2019–20 produced the lowest default risk cohorts
- 2022–24 show a gradual uptick in default rates
- Model PD tracks vintage trends reasonably well

Highest risk cell:
Partner E × 2024 = 13.2%
default rate — flagged for review.

Default Rate Heatmap — Vintage Year × Lending Partner

Darker cells = higher observed default rate | Partner E × 2024 is the highest risk concentration



Limitations & Next Steps

Honest assessment after one week of development

Known Limitations

- PD scores overstate raw probabilities (calibration step needed)
- Partner data is simulated — model cannot meaningfully differentiate partners yet
- LGD assumed at 50% — needs VersaBank actual recovery data
- Loan Term and some Loan Purpose variables show no predictive power
- Pseudo $R^2 = 0.119$ — 12% of default variation explained
- Origination dates simulated — real RPP transaction dates needed

Priority Next Steps

- Replace simulated partner data with real RPP originator IDs
- Apply Platt Scaling to calibrate PD scores to actual probabilities
- Source VersaBank recovery data for empirical LGD estimation
- Add macroeconomic variables (unemployment, BoC rate, HPI)
- Build a survival model (Cox PH) to model timing of default
- Automate pipeline for monthly re-run as new RPP data arrives

The framework, methodology, and code are all in place. Closing these gaps is a data and time problem — not a modelling problem.

Summary

What this framework delivers — right now

- 01 End-to-end pipeline from raw loan data to holdback recommendation
- 02 Statistically validated model (ROC-AUC 0.7531) with regulatory-grade output
- 03 $EL = PD \times EAD \times LGD$ holdback framework aligned to RPP structure
- 04 Ten years of vintage analysis identifying Partner E / 2024 as highest risk
- 05 Fully documented, reproducible code ready for audit or handoff